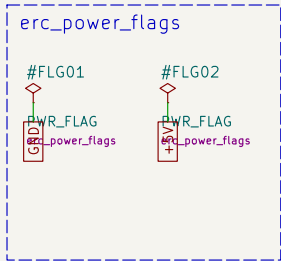
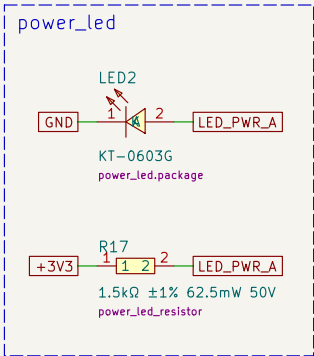
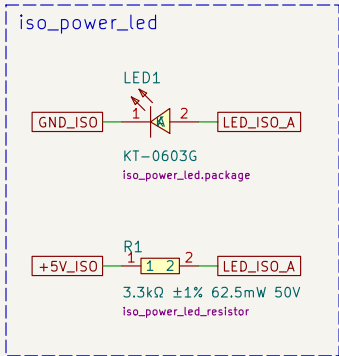
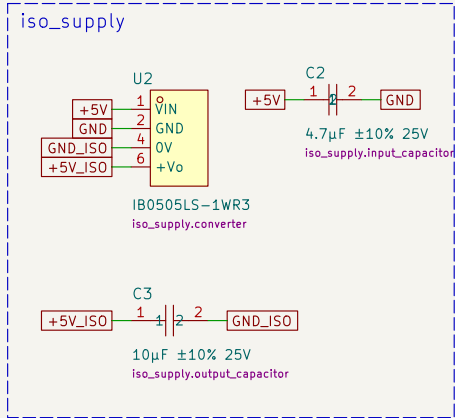
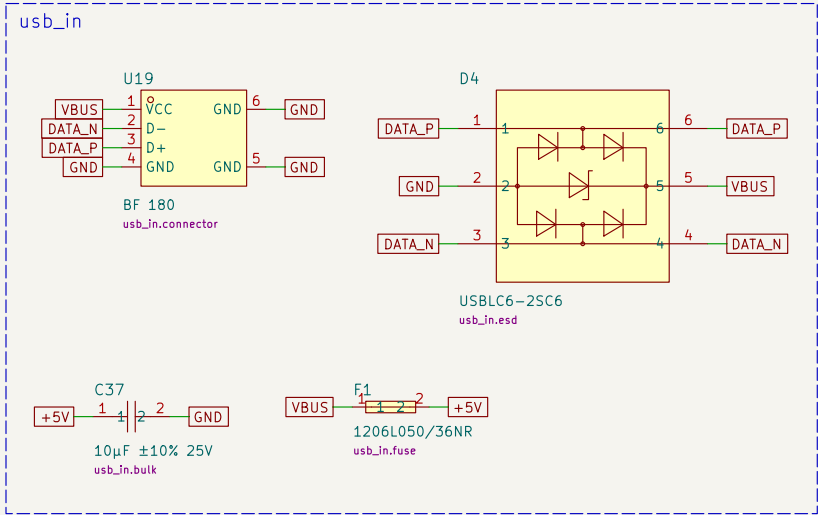
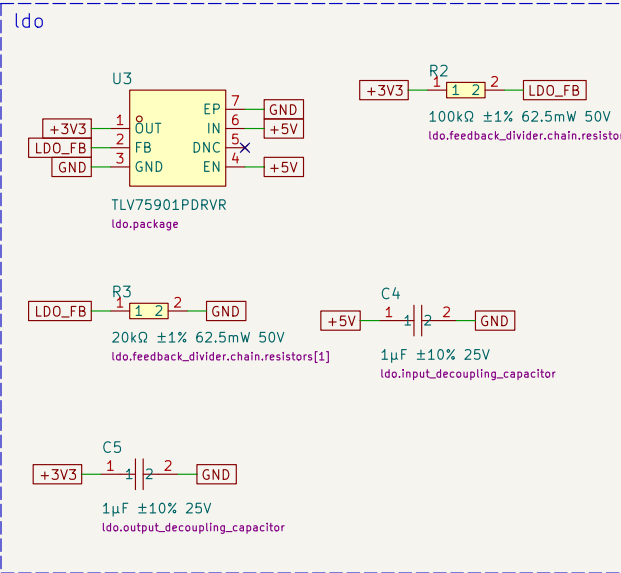


	1	2	3	4	5	6	7	8
A	Solartron 7075 USB interface RP2354A USB interface for the Solartron 7075 DVM + 70754 Parallel BCD Interface (SKB, 50-way Cannon). Everything to/from the meter is opto-isolated; only SCK, MOSI, LATCH, OE_N, MISO and DRDY cross the barrier. Nets are joined by name (global labels). GND (USB side) and GND_ISO (DVM side) are never joined.							
B		2. USB input and power File: power_usb.kicad_sch		USB-B -> 500 mA PTC + USBLC6 ESD -> +5V. +5V -> TLV75901 (100k/20k feedback) -> +3V3. +5V -> IB0505LS-1WR3 isolated DC-DC -> +5V_ISO / GND_ISO (DVM side).				
C		3. RP2354A File: mcu.kicad_sch		RP2354A (2 MB in-package flash). Core regulator per RP2350 datasheet: 3.3 uH polarity-marked inductor, 4.7 uF CIN/COUT, 33R + 4.7 uF on VREG_AVDD. GPIO16 MISO, GPIO17 LATCH, GPIO18 SCK, GPIO19 MOSI, GPIO20 OE_N, GPIO21 DRDY, GPIO25 status LED.				
D		4. Isolation barrier File: isolation.kicad_sch		Six non-inverting TLP2361 channels. The LED cathode is driven by the source, the anode is fed through 330R (3.3 V side) or 680R (5 V side), ~4.5 mA. Undriven inputs leave the outputs HIGH (OE_N_ISO high = 595 outputs Hi-Z).				
E		5. Solartron SKB interface File: dvm.kicad_sch		SKB pins 1-36 -> 5x 74HCT165 (frame bit k = SKB pin k+1, then 1010 signature). 2x 74HCT595 -> SKB 38, 40-50; 2N7002 contact-closure SAMPLE on SKB 39. SKB 37 = DVM earth / logic 0 = GND_ISO. See README.md for the bit map.				
F								

USB-B -> 500 mA PTC + USBLC6 ESD -> +5V.
+5V -> TLV75901 (100k/20k feedback) -> +3V3.
+5V -> IB0505LS-1WR3 isolated DC-DC -> +5V_ISO / GND_ISO (DVM side).



The .ato files are the source of truth – do not edit this schematic
Auto-generated from the atopile netlist by scripts/export_schematic.py
atopile design (.//*.ato)
Sheet: /2. USB input and power/
File: power_usb.kicad_sch

Title: Solartron 7075 USB interface – USB input and power

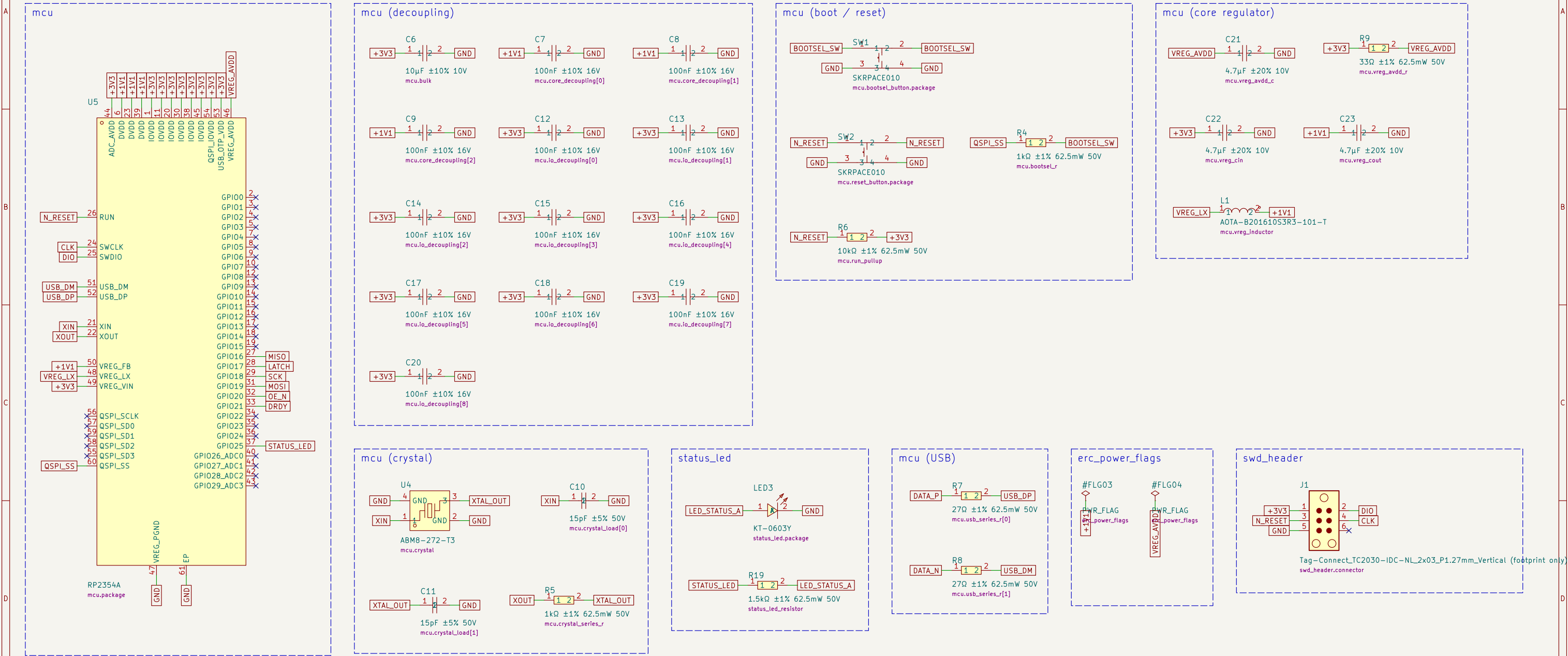
Size: A4
KiCad E.D.A. 10.0.6

Date: 2026-10-07

Rev: A

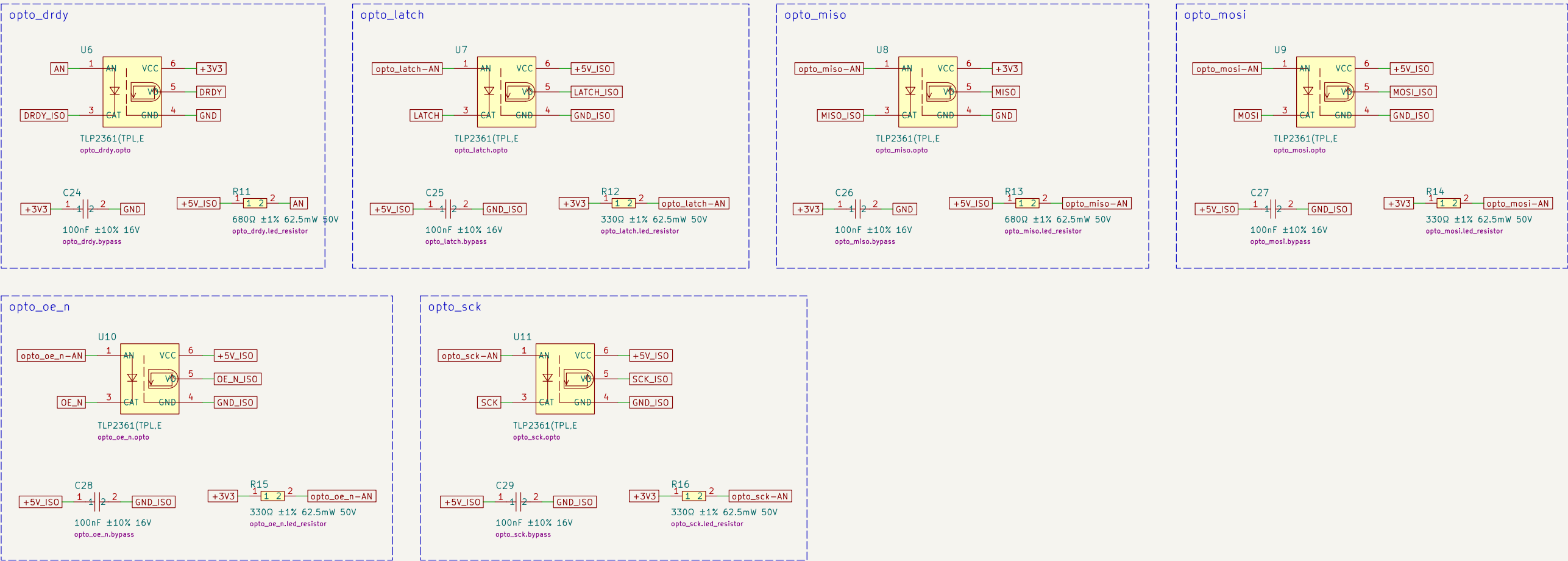
Id: 2/5

RP2354A (2 MB in-package flash). Core regulator per RP2350 datasheet:
3.3 uH polarity-marked inductor, 4.7 uF CIN/COUT, 33R + 4.7 uF on VREG_AVDD.
GPIO16 MISO, GPIO17 LATCH, GPIO18 SCK, GPIO19 MOSI, GPIO20 OE_N, GPIO21 DRDY, GPIO25 status LED.



The .ato files are the source of truth – do not edit this schematic
Auto-generated from the atopile netlist by scripts/export_schematic.py
atopile design (.//*.ato)
Sheet: /3. RP2354A/
File: mcu.kicad_sch

Six non-inverting TLP2361 channels. The LED cathode is driven by the source, the anode is fed through 330R (3.3 V side) or 680R (5 V side), ~4.5 mA. Undriven inputs leave the outputs HIGH (OE_N_ISO high = 595 outputs Hi-Z).



The .ato files are the source of truth – do not edit this schematic
Auto-generated from the atopile netlist by scripts/export_schematic.py
atopile design (../*.ato)
Sheet: /4. Isolation barrier/
File: isolation.kicad_sch

Title: Solartron 7075 USB interface – Isolation barrier

Size: A3	Date: 2026-10-07	Rev: A
KiCad E.D.A. 10.0.6		Id: 4/5

SKB pins 1–36 -> 5x 74HCT165 (frame bit k = SKB pin k+1, then 1010 signature).
2x 74HCT595 -> SKB 38, 40–50; 2N7002 contact-closure SAMPLE on SKB 39.
SKB 37 = DVM earth / logic 0 = GND_ISO. See README.md for the bit map.

